Military Institute of Science and Technology
Department of Computer Science and Engineering

Peripheral/Computer Connections (Part 2)

Course Name: Computer Interfacing
Course Code: CSE-405

Book Reference

- Computer Peripherals (3rd Edition) - Cook and White; Butterworth- Heinemann (1995)

Block Data Transfer

- Processors have instructions for moving **blocks of data** which may be used for I/O.
- **CPU has to synchronize** its block moves with the peripheral and is unable to performs any other functions while the block moves is in progress.
- **Only suited for fast transfers** where the CPU and the peripheral speeds are reasonably well matched.
- It is not commonly used.

Direct Memory Access (DMA)

- It is better to avoid using the CPU completely for I/O transfer.
- A special hardware called **DMA controller** can be used to transfer data between memory and I/O devices.
- When a peripheral indicates that it is ready for data transfer
  - The **DMA unit gains the control of the Bus**.
  - Places **appropriate address and control signals** on it to make the transfer.
  - **Releases the bus**.
- This action of taking over the bus for a period and **executing a memory access cycle** instead of the CPU doing so is known as *cycle stealing.*

Parallel Interface: SCSI

Parallel Interface: SCSI

- **SCSI (Small Computer System Interface)** is a **set of standards** for physically **connecting and transferring data between computers and peripheral devices especially storage devices.**
- Used to connect computers to disk drives and other peripherals (generally those needing a high data rate).
- A complete communication protocol and bus standard
- **8 devices can be supported on the SCSI bus**
  - Each device can have 8 logical units
  - Each logical unit contains a unique Logical Unit Number (LUN)
  - Each logical unit can have 256 logical sub units
    - Thus a very large total number of possible devices

Parallel Interface: SCSI

- Each device can be either an **initiator**, a **target** or both.

Initiator:

- A SCSI device that requests an operation to be performed **by another SCSI device**.
- Initiate a SCSI session.
- Does not provide any Logical Unit Number.

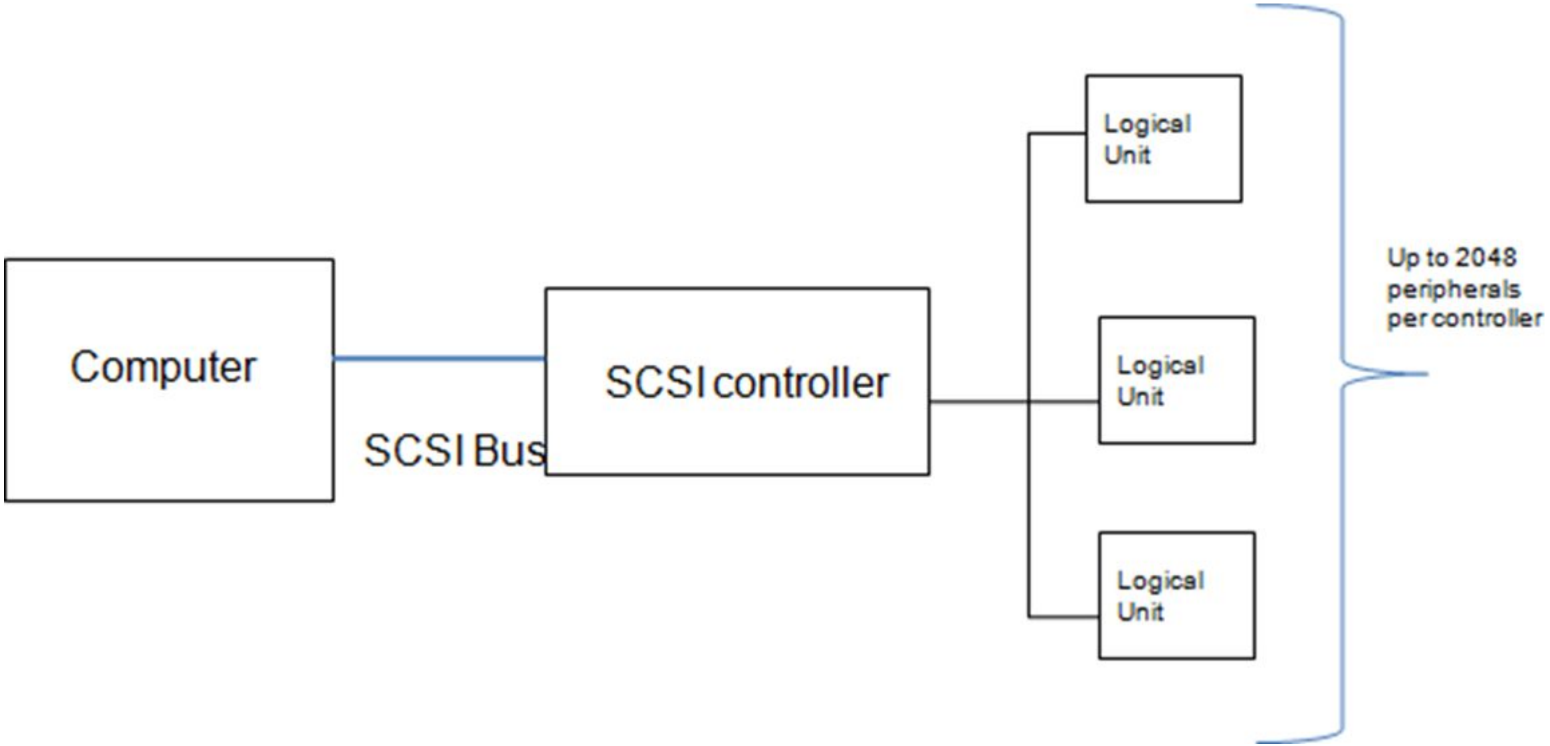
Target:

- A SCSI device that performs an operation as requested by an initiator.
- Waits for initiators' commands and provides required input/output data transfers.
- Provide one or more Logical Unit Number.

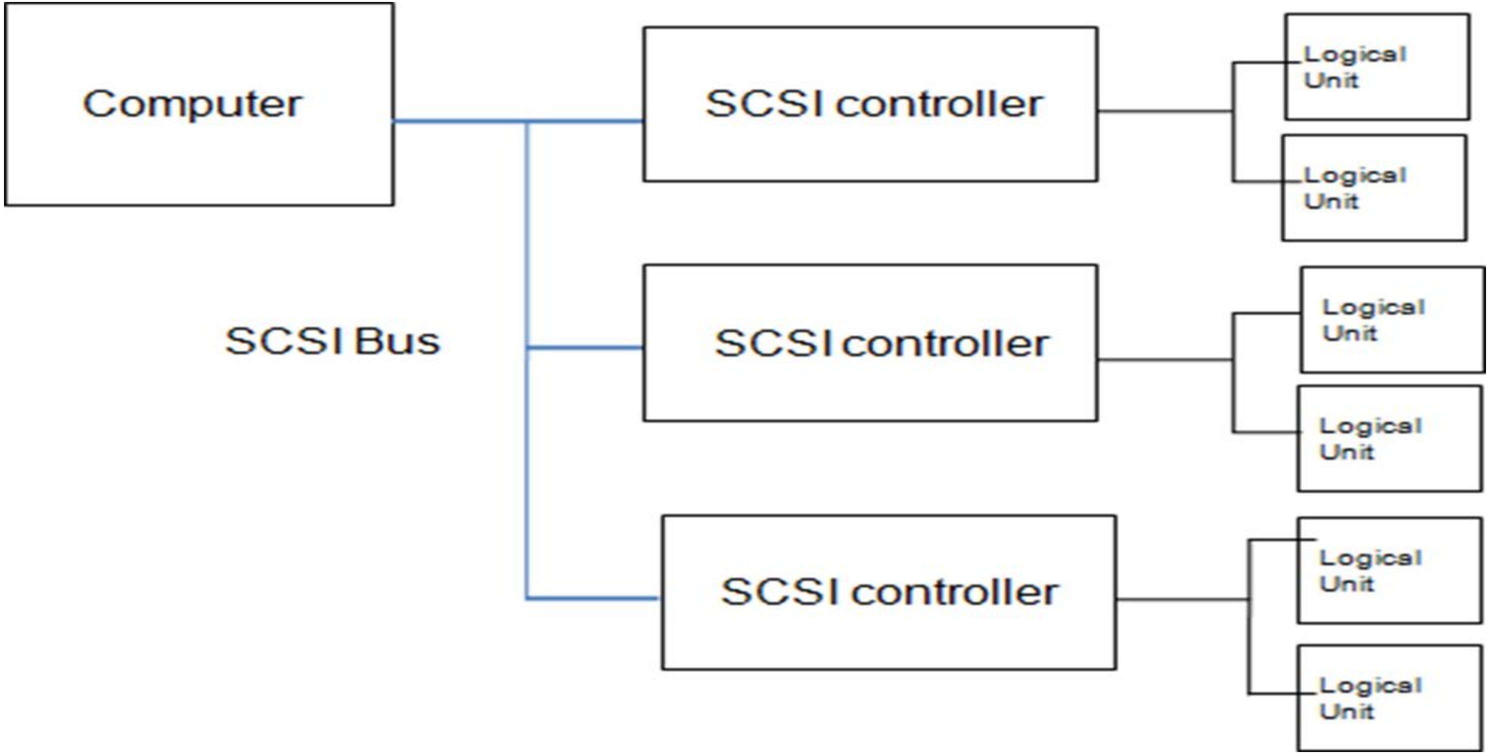
Parallel Interface: SCSI

- To be useful SCSI bus **must have at least one initiator and one target**
  - Can have multiple initiator and/or target
- Data is transferred over an **8 bit bidirectional bus**
  - **Optional parity line to check the error** of transmission
- **Data and control lines are implemented** using **one wire each together with common ground in single-ended SCSI**
  - With pair of wires for each line in differential SCSI
- When an **end user sends a request**, the **operating system sends the SCSI commands to the SCSI controller**, which then sends it to the storage device.
- **SCSI controller** (embedded on motherboard) is a **host bus adapter** or a chip that allows a SCSI storage device to communicate with operating system across a SCSI bus.

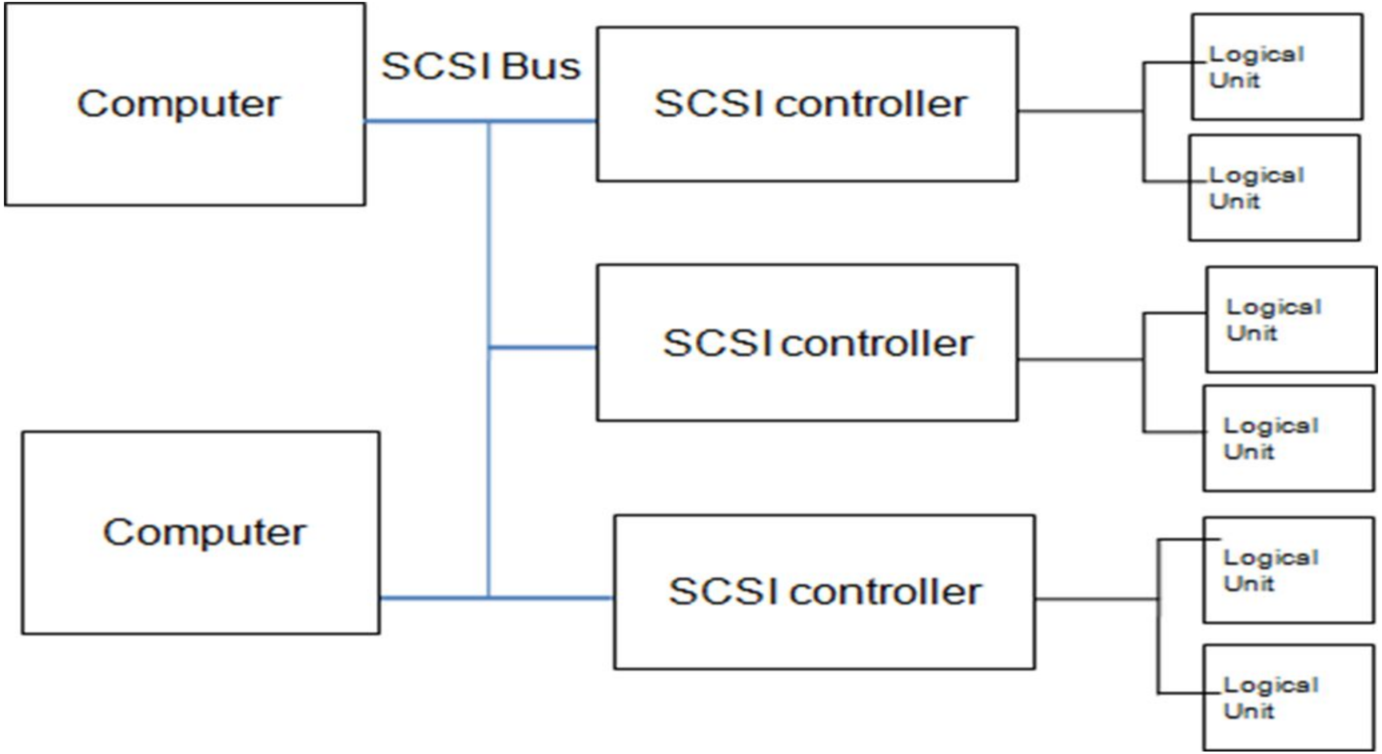
One Initiator, One Target



One Initiator, Multiple Targets



Multiple Initiators, Multiple Targets



Protocol for Using SCSI Bus

- Phase 1 : Arbitration
- Phase 2 : Selection
- Phase 3 : Information Transfer
- Phase 4 : Bus free
- Phase 5 : Reselection

➢ Several commands can be received before the first is completed

➢ Devices can queue requests

Protocol for Using SCSI Bus

Phase 1: Arbitration

- Allows one SCSI device (here initiator) to gain control of the SCSI bus so that it can initiate or resume an I/O process.
- The **SCSI device shall first wait for the BUS FREE phase** to occur

Phase 2: Selection

- The SELECTION phase **allows an initiator to select a target** for the purpose of initiating some target function.

Phase 3: Information transfer

- The COMMAND, DATA, STATUS, and MESSAGE phases are combined to transfer data or control information.

Protocol for Using SCSI Bus

Phase 4: Bus Free

- BUS FREE phase is entered when a **target release the BUSY signal**
- The BUS FREE phase indicates that **there is no current I/O process and the SCSI bus is available for a new connection.**

Phase 5: Reselection

- When the target is ready to respond to the given command then it can re-establish contact with the initiator in Reselection phase.
- Connection that started by the initiator but was **suspended by the target**.

SCSI: Mathematical Problem-1

Initiator	Target	Arbitration time of initiator (sec)	Task time of target given by initiator (sec)
2	1	4	8.4
1	3	5.1	5
1	2	6.4	5.8
1	3	8	3.5
2	2	8.2	4.9

Suppose, 2 computers are sharing a common SCSI bus among them. 3 SCSI controllers are connected with the same bus. Now solve the following scenario and find out the value of t at which all the controllers and bus will become free after executing all the tasks given by the all the computers.

Consider ,data transfer rate between all components = 1.7 sec

And arbitration +selection/reselection time= 2 sec

SCSI: Mathematical Problem-2

Initiator	Target	Arbitration time of initiator (sec)	Task time of target given by initiator (sec)
2	3	1	10
1	2	3	8
1	3	6	4
3	1	8	20
3	3	12	9

Suppose, 3 computers are sharing a common SCSI bus among them. 3 SCSI controllers are connected with the same bus. Now solve the following scenario and find out the value of t at which all the controllers and bus will become free after executing all the tasks given by the all the computers.

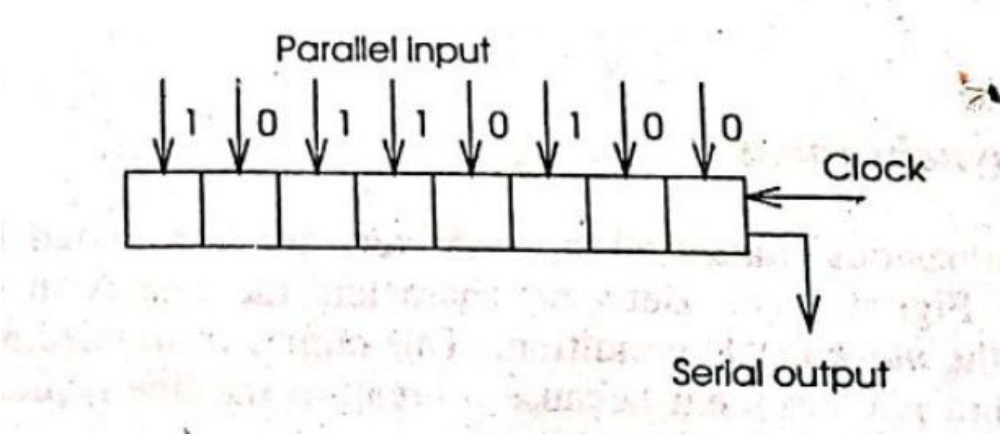
Consider ,data transfer rate between all components = 2 sec

And arbitration +selection/reselection time= 1 sec

Serial Interface

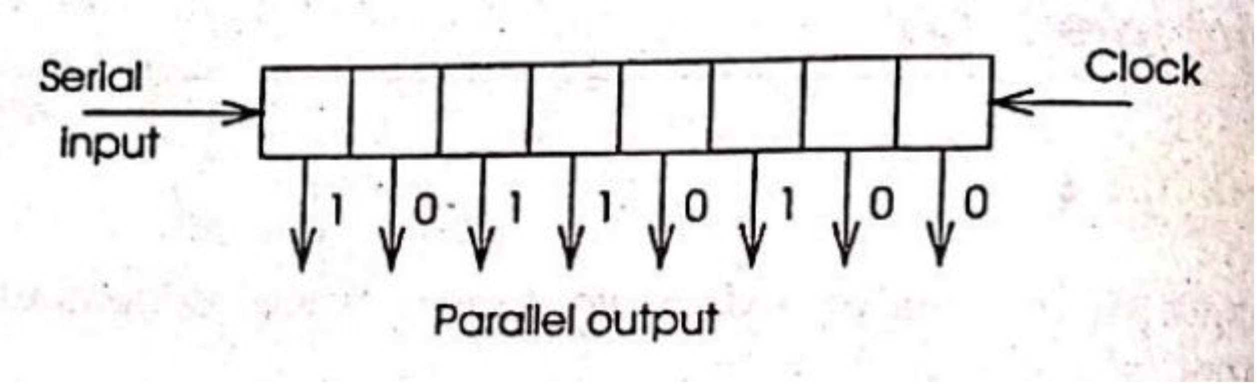
Parallel to Serial Interface

- Simplest convertor is a **shift register**
- Parallel data word is loaded to into a shift register
- A pulse on the clock input shift the data by 1 bit. And output 1 bit.



Serial to Parallel Interface

- Sequence of n clock pulse caused the input to propagate along the shift register
  - Until it is all available in parallel
  - First bit shifted all the way through shift register and appears at the right end.
- **The clock in the receiver must operates with the same timing as the transmitter.**



Asynchronous Transmission

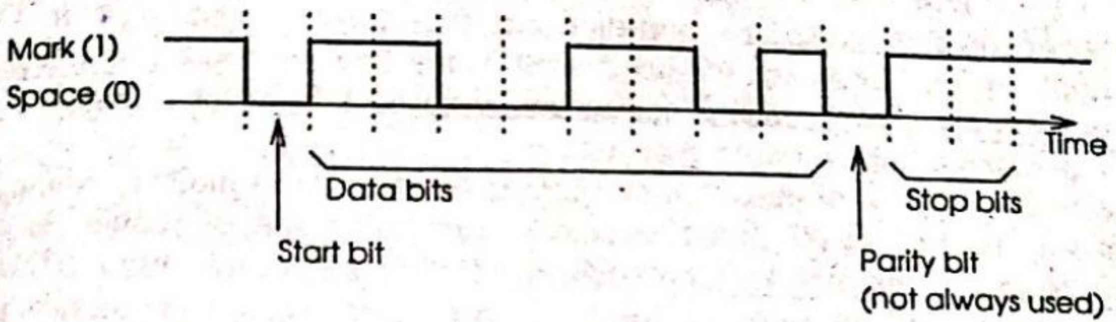


Figure 2.16 *Asynchronous character format*

Asynchronous Transmission

- Between characters the lines are is in the **idle state** which is the **'mark' or '1'** condition
- The first bit of the character, the **'start bit'** is in the **'space' or '0' condition**
- The 1 to 0 transition is used to start the receiver clock and indicates that a character is arriving
- The **first bit of data can not be the start bit**
- Because if t**he first bit is 1 it will be indistinguishable from the idle condition**
- For the data part of the character 5,6,7 or 8 data bits are transmitted with least significant bit first
- There may be a **parity bit to allow transmission error** to be detected
- Finally, the line returns to a idle state fo**r 1, or 2 stop bits which allow the receiver to** settle ready for the next character
- Also known as 'stop-start' transmission

Asynchronous Transmission

- Irregular intervals may occur between character
- The s**ender and the receiver are not synchronized at the character level**
- Hence known as asynchronous transmission

Synchronous Transmission

- Supply a **clock signal** and **a data signal**
- Transmission may be along a **pair of line, one for clock signal and one for data**
- Clock may be merged with data stream into a single line
- Two advantages:
  1. **No time wasted** for non-data bits
  2. **Less margin of error** is required to ensure correct recovery of data- increase acceptable **transmission speed**
- **Use a special sequence of data known as protocols**
- One protocol: HDLC (High Level Data Link Control)

Synchronous Transmission : HDLC format

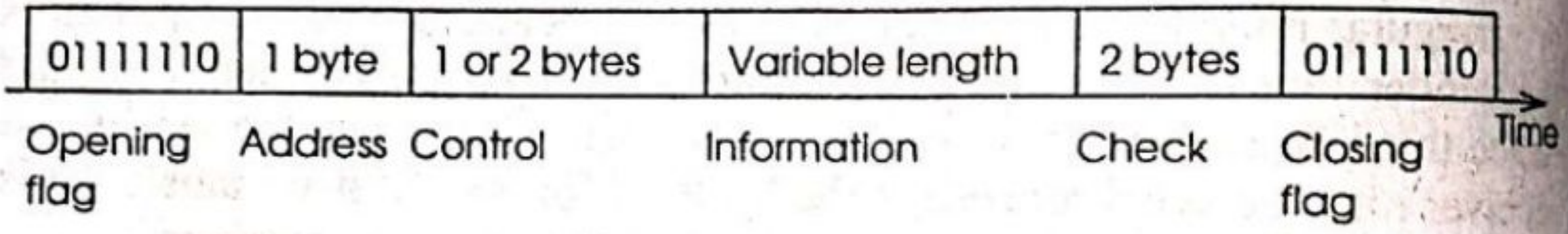


Figure 2.17 *HDLC synchronous format*

Synchronous Transmission: Bit stuffing

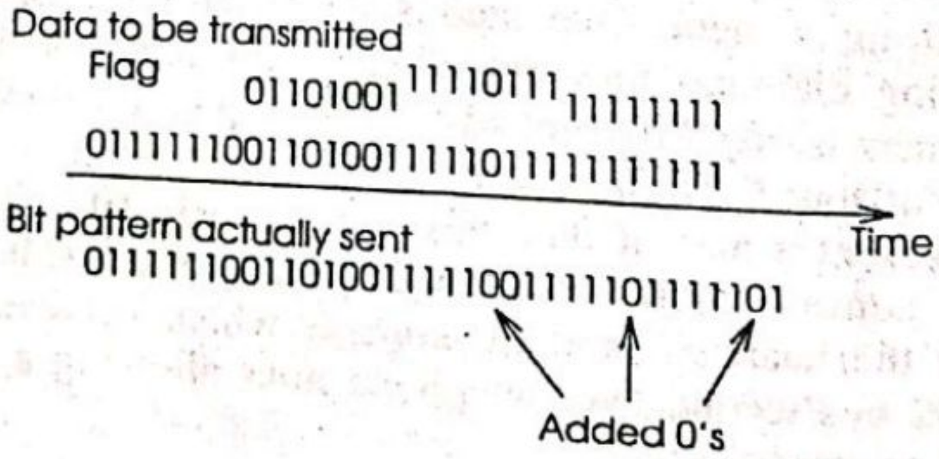


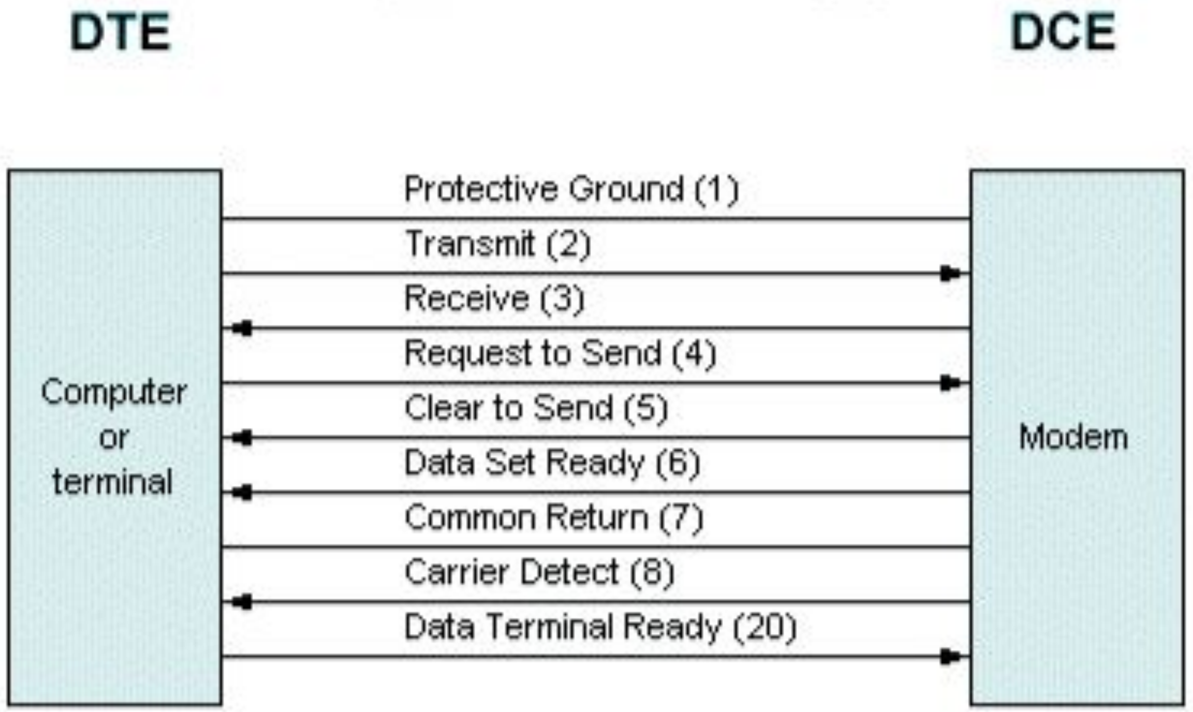
Figure 2.18 *Example of bit stuffing*

The RS232 Standard

- A standard used for serial data transmission.
- Communication is assumed to be between **Data Terminal Equipment (DTE) and Data Communication Equipment (DCE).**
- DTE can be a computer or peripheral, DCE is another device such as a modem **that connects the DTE to a communication network**.
- **Standard telephone** line can be used due to its availability.

The RS232 Standard Protocol

- When the terminal is powered on, it performs **a self-checking and asserts DTR (Data Terminal Ready) signal** to tell the modem that it is ready.
- When modem is ready, **it sends DSR (Data Set Ready) signal to the terminal**.
- Then the modem dials the computer.
- When terminal is ready to send, **it asserts RTS (Request to send) signal**.
- Modem asserts **CD (Carrier Detect) signal** to indicate that **it has established contact with computer**.
- When modem is ready, it asserts **CTS (Clear to Send) signal**.
- Terminal sends data.
- When all data has been sent, **terminal raises**, **RTS signal high**.
- Modem raises **CTS signal high**.


DTE
DCE
Computer or terminal
Modem
Protective Ground (1)
Transmit (2)
Receive (3)
Request to Send (4)
Clear to Send (5)
Data Set Ready (6)
Common Return (7)
Carrier Detect (8)
Data Terminal Ready (20)

Thank You